# Global Facilities - Design & Construction Standard - CSA - Pile Load Test Specifications

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|---|---|---|---|
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**ECN History** **Submit a Change Request**

## 1.0 Purpose

1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Pile Load Test as part of any green and/or brown field construction projects contract requirement.

1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.

1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.

1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:

- 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
- 1.4.2 Environmental, Health & Safety Standards
- 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
- 1.4.4 Security Protocol

1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.

1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

## 2.0 Scope

2.1 This document for pile load test specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

## 3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

| Term/Acronym | Definition / Description |
|---|---|
| FCT | Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs |
| GFTT | Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards |
| GFC | Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities |
| Global Facilities Central Team | This is used interchangeably with Innovation & Integration Team |
| Global Facilities Construction Team | This is used interchangeably with Global Construction Team |
| Global Facilities Operations Support Team | This is used interchangeably with Global Facilities Operations Team |
| Global Facilities Planning Team | This is used interchangeably with Global Facilities Line Planning Team |
| AE | External Architectural & Engineering Consultants |
| DPT | Micron's Designated Project Team |
| GFCT | Micron's Global Facilities Construction Team |
| EHS | Micron's Environmental Health & Safety Team |
| QP/PE | External Qualified Person (professional architect/engineer) |
| SME | Micron's Subject Matter Expert |
| TM | Micron's Team Members |

## 4.0 General

4.1 Load tests shall be carried out using load cells and in accordance with or equivalent local code requirements,

**BS EN 1997-1** Geotechnical design - Part 1: General rules,
**BS EN 1997-2** Geotechnical design - Part 2: Ground investigation and testing.
The test load shall be measured by a calibrated load gauge as well as calibrated pressure gauges.

4.2 The Contractor shall supply all labour and materials and all other equipment necessary for the performance, recording and measurements of the test loads and settlement including the supply and placing in position kentledge used in the tests and subsequent removal of test load, dismantling of staging as well as recording instruments, etc. for the proper completion of the tests.

4.3 Throughout the duration and operation of the test loading, the Contractor shall place competent person to operate, watch and record the test.

4.4 The load test shall not be implemented until at least 7 days have elapsed after driving has been completed or until concrete has reached its specified strength for cast-in-situ bored piles.

4.5 Installation of pile near the pile load testing area shall be suspended during the testing period.

4.6 The scales, hydraulic jacks, load cells, and load and pressure gauges shall be approved by the Engineer before any load test is carried out. They shall be calibrated and tested by an approved accredited laboratory before each use. The calibration charts and test certificate shall be submitted to the Engineer before any load tests are carried out. The Contractor shall take all precautionary measures as instructed by the Engineer to prevent any interference with the apparatus during the period of load test.

4.7 The entire test area must be sheltered from direct sunlight, wind and rain and sufficiently lighted to facilitate taking readings.

4.8 This document applies to all Micron Facilities employees, contractors and vendors at all Micron sites.

## 5.0 Preliminary Test Pile for Ultimate Load Test

5.1 The contractor shall provide test piles for ultimate load tests where specified before the installation of the working piles. The test piles shall be installed separately from the working piles and in positions determined by the Engineer. Test piles shall be installed in an approved manner to a predetermined depth and /or set. Load tests on bored piles shall be instrumented and carried out in accordance with 'Instrumented Load Test' specifications.

5.2 The test pile is to be of the same materials and dimension as the working piles and installed or driven with the same type of plant and by the same method and procedure, so as to ensure that the behaviour under test loads are comparable with those of the working piles.

5.3 The performance of the test piles during installation and later under test load will be used to determine the bearing capacities of the working piles and driving criteria.

5.4 Ultimate load test shall be done on preliminary test piles only, and not on working piles

## 6.0 Ultimate Load Test of Working Piles

6.1 The pile under ultimate load test shall be subject to a test load up to three times the working load of the pile.

6.2 Failure Load of ultimate load test is defined as the load at which settlement continues without further load increment.

## 7.0 Standard Load Test of Working Piles

Standard load tests to 2 times the Working Loads shall be conducted on working piles to be selected by the Engineer as the work progresses.

## 8.0 Method of Loading

8.1 In both Ultimate Load Test and Standard Load Test, incremental or decremental test loads shall be applied by direct loading on the piles with kentledge. The type of kentledge used can be of concrete blocks, piles, metal ingots, etc., but must be of uniform regular sizes such that the weights can be easily calculated. The total kentledge weight shall be at least 1.2 times the full test load. The pile cap for load tests, if necessary, shall be designed and built with rapid hardening cement by the Contractor and to the approval of the Engineer. The cost of building and demolishing such pile caps shall be borne by the Contractor.

8.2 The Contractor shall provide temporary supports at the corners of the platform for the kentledge so that in building up the test loads, bending moments on the pile will not be caused when handling the kentledge and also to prevent accidents due to sudden settlements. The cost so involved shall be borne by the Contractor.

8.3 Once any load test is carried out, it shall be continued until the whole test load is completed and all measurements of settlements taken are recorded.

8.4 The procedure of Load Tests, and application and removal of test loads shall be in accordance with:

Schedule 1 – Standard Working Loading Procedure

Schedule 2 – Ultimate Load Test Loading Procedure

8.5 All loading and unloading operation shall take place during the day and in the presence of the Engineer’s Representative.

## 9.0 Measurement of Settlements

9.1 The method of measurement of settlement shall be by a primary system of 2 dial gauges set at two opposite corners of pile and a secondary system of using level and staff. Reading shall be taken with both systems at appropriate intervals. Dial gauges shall have a 50mm travel and be accurate to 0.1mm.

9.2 The level and the scale of the staff shall be chosen to enable readings to be made to an accuracy of 0.5mm. A scale attached to the pile or pile caps may be used instead of a leveling staff. A datum shall be established on a permanent object or other well-founded structure and shall be situated so that only one setting of the level is needed. The datum marker shall be duplicated and can be re-established in case it is inadvertently demolished. The datum must not be affected by the test loading or other operation on the site.

9.3 Settlement reading s shall be taken before and after application of each increment or decrement of load. Reading shall be recorded in a manner approved by the Engineer.

9.4 The test load shall be measured by a calibrated load gauge as well as a calibrated pressure gauge in the hydraulic system.

9.5 The Contractor shall erect a temporary shelter to protect the instrument against disturbances or other approved arrangements to prevent disturbances or theft of the instrument.

## 10.0 Test Reports

10.1 The report shall contain the following:

(a) Pile designation, date completed, weather condition, pile length, pile size, water table, set, hammer data, length of casing if any, method of casting of bored pile and level of pile bottom.

(b) Description of the apparatus used for testing, loading system and procedure for measuring settlement.

(c) Filed data.

(d) Time-settlement curve.

(e) Load-settlement curve shall show the rebound or residual settlement.

(f) Remarks explaining unusual events or data and movement of piles.

## 11.0 Failure of Ultimate Load Test Pile

11.1 Ultimate Load Test Pile under loading test shall be deemed to have failed when one of the following observations is recorded:

(a) Any portion of the pile is found to have cracked, crumbled, distorted from its original shape or alignment deflected from its original position and the like.

(b) The loading test is unable to be completed due to instability of the kentledge or tampering of the levels or scales or reference datum by the Contractor.

(c) Any portion of the pile cap exhibit any sign of failure as considered by the Engineer.

(d) Pile under load test continues to settle without further load increment.

## 12.0 Failure of Standard Load Test Piles

12.1 The Engineer's interpretation and conclusion arrived at on the test results shall be final. When the pile test has been carried out as described in the preceding clauses, the pile so tested shall be deemed to have failed to meet the requirements of this Specification, if:

(a) The maximum settlement under One 9times the nominal working load of pile exceeds 10mm.

(b) The settlement at column load plus 2 times negative skin friction on pile, when applicable, exceed 10mm.

For steel piles only, the maximum settlement under full test load given in the above may be increased by not more than 0.15mm per meter length of the pile to allow for the elastic shortening of the piles.

(c) The maximum settlement under 2 times the nominal working load of pile exceeds 25mm.

(d) The permanent or residual settlement after removal of the test load exceeds 13mm.

(e) The load test is unable to be completed due to instability of the kentledge, failure of pile cap, or tampering of the levels or scales or reference datum by the Contractor or continuous settlement of pile under load test.

(f) Any portion of the pile is found to have cracked, crumbled, distorted from its original shape or alignment deflected from its original position and the like.

12.2 Any pile which fails under Standard Loading Test shall be replaced by one or more piles at the Contractor's expense as directed by the Engineer and in addition, the Contractor shall at his own expense carry out two further loading tests on piles selected by the Engineer. The Contractor shall also bear the extra cost which may be necessary from a re-design of the pile cap, ground beam, etc. because of the failed pile.

## 13.0 Measurement of Settlements

Unless otherwise specified, minimum number and type of load tests on piles shall be:

(A) For projects with building structures of 10-storey or more:

**Table 1**

| Type of Load Test | Pile Test Schedule |
|---|---|
| (a) Ultimate load test on preliminary pile (instrumented for bored piles) (3 times working load) | 1 number or 0.5% of the total piles whichever is greater |
| (b) Working Load Test (static) | 2 numbers or 1% of working piles installed or 1 for every 50 meters length of proposed building, whichever is greater |
| (c) Non-Destructive Integrity Tests<br>❖ For precast RC piles : Pile Integrity Tests (PIT)<br>❖ For RC bored piles : Pile Dynamic Analyser (PDA) Tests | 2 numbers or 2% of working piles installed, whichever is greater |

(B) For projects with building structures below 10-storey:

**Table 2**

| Type of Load Test | Pile Test Schedule |
|---|---|
| (a) Ultimate load test on preliminary pile (instrumented for bored piles) (3 times working load) | 1 number if total no of piles less than 200 and 2 numbers if total no of piles exceed 200 |
| (b) Working load test (static) | 0.75% of total number of piles (minimum 1 number) |
| (c) Non-Destructive Integrity Tests<br>❖ For precast RC piles : Pile Integrity Tests (PIT)<br>❖ For RC bored piles : Pile Dynamic Analyser (PDA) Tests | 2 numbers or 2% of working piles installed, whichever is greater |

## 14.0 Abandonment of Pile Test

A test shall be abandoned if it has to be discontinued due to:

(a) pre-jacking or pre-loading before the commencement of the test

(b) improper setting of datum

(c) disturbed or unstable bench marks or scales

(d) faulty jack or gauges

(e) failure of pile cap or instability of the kentledge

(f) failure of the pile materials

(g) failure of concrete to reach the design compressive strength

(h) stoppage due to breakdown of installation or driving equipment

(i) the scales and/or measuring instruments used are found to be defective or have been tampered with

The cost of the abandoned test, replacement of piles and further tests shall be borne by the Contractor.

## 15.0 Dynamic Pile Testing (PDA Tests)

15.1 Where specified, dynamic pile testing shall be carried out by an approved Specialist to evaluate the performance and integrity of installed reinforced concrete piles. The analysis and results of the dynamic pile testing carried out shall be correlated with results of static load tests performed to yield the final load/settlement curves and pile capacities.

15.2 The Contractor shall provide all necessary machinery, equipment and instrumentation for the carrying out of dynamic pile testing and analysis. These shall include piling rigs and hammers, gauges, accelerometers, strain transducers, special equipment and computers. Fieldworks shall include the necessary building-up of pile-heads and setting up for the tests.

15.3 Dynamic pile testing shall be carried out with the use of Pile Driving Analyzer (PDA). The field results shall be further analyzed using the computer program CAPWAP (Case Pile Wave Analysis Program – Continuous Version).

15.4 The pile's mobilized static bearing capacity shall be evaluated from field testing using the CASE METHOD of analysis.

15.5 Value of mobilized total static resistance and skin resistances shall be computed from the CAPWAP analysis.

The analysis shall also yield simulated static load test results on total pile top settlements at pile working load and twice working load. Settlement results shall be plotted on Load/Settlement curves.

15.6 Within 3 days from the completion of field testing, the Contractor shall furnish to the Engineer, 2 copies of Report certified by a Professional Engineer, and containing details of field testing carried out, analysis and dynamic pile testing results.

## 16.0 Sonic Echo Tests

For cast-in-situ bored piles, pile integrity tests using sonic echo or equivalent method shall be conducted on not less than 2% of the total number of piles installed.

## 17.0 Completion of Load Test

The Contractor shall allow in his Tender costs for the demolition of all pile caps constructed for the carrying out of loading tests of piles and also stripping of test piles to pile cut-off levels.

STANDARD WORKING LOAD TEST LOADING PROCEDURE

(Two times the specified working load)

**Table 3**

| Load Cycle No. | Day & Time | Load (% of WL) | Minimum Holding Time |
|---|---|---|---|
| 1 | 1st Day | 0.0% | |
| | 10.00 am | 12.5% | 30 min. |
| | 10.30 am | 25.0% | 30 min. |
| | 11.00 am | 37.5% | 30 min. |
| | 11.30 am | 50.0% | 30 min. |
| | 12.00 noon | 62.5% | 30 min. |
| | 12.30 pm | 75.0% | 30 min. |
| | 1.00 pm | 87.5% | 30 min. |
| | 1.30 pm | 100.0% | 24 hrs. |
| | | | |
| | 2nd Day | | |
| | 1.30 pm | 87.5% | 30 min. |
| | 2.00 pm | 75.0% | 30 min. |
| | 2.30 pm | 62.5% | 30 min. |
| | 3.00 pm | 50.0% | 30 min. |
| | 3.30 pm | 37.5% | 30 min. |
| | 4.00 pm | 25.0% | 30 min. |
| | 4.30 pm | 12.5% | 30 min. |
| | 5.00 pm | 0.0% | 60 min. |
| | | | |
| 2 | 3rd Day | 0.0% | |
| | 8.30 am | 25.0% | 30 min. |
| | 9.00 am | 50.0% | 30 min. |
| | 9.30am | 75.0% | 30 min. |
| | 10.00 am | 100.0% | 30 min. |
| | 10.30 am | 112.5% | 30 min. |
| | 11.00 am | 125.0% | 30 min. |
| | 11.30 am | 137.5% | 30 min. |
| | 12.00 noon | 150.0% | 30 min. |
| | 12.30 pm | 162.5% | 30 min. |
| | 1.00 pm | 175.0% | 30 min. |
| | 1.30 pm | 187.5% | 30 min. |
| | 2.00 pm | 200.0% | 24 hrs. |
| | | | |
| | 4th Day | | |
| | 2.00 pm | 175% | 30 min. |
| | 2.30 pm | 150.00% | 30 min. |
| | 3.00 pm | 125% | 30 min. |
| | 3.30 pm | 100.00% | 30 min. |
| | 4.00 pm | 75% | 30 min. |
| | 4.30 pm | 50.00% | 30 min. |
| | 5.00 pm | 25% | 30 min. |
| | 5.30 pm | 0% | 24 hrs. |

| Load Cycle No. | Day & Time | Load (% of WL) | Minimum Holding Time |
|---|---|---|---|
| 2 | 5th Day | | |
| | 8.30 am | 0.0% | |
| | Final rebound reading after 24 hrs from zero load. | | |

Following each application of load, hold the load for not less than the period shown or until the rate of settlement is less than 0.1mm/hr, whichever is later.

For any period when the load is constant, record time and settlement immediately on reaching the load and at 15-min. intervals for the 1st hour, at 30-min. intervals between the 1st and 4th hour and at 1-hr intervals between the 4th and 24th hour after the application of the increment of the load.

**Table 4**

**ULTIMATE LOAD TEST LOADING PROCEDURE**

(Three times the specified working load)

| Load Cycle No. | Day & Time | Load (% of WL) | Minimum Holding Time |
|---|---|---|---|
| 1 | 1st Day | 0.0% | |
| | 10.00 am | 12.5% | 30 min. |
| | 10.30 am | 25.0% | 30 min. |
| | 11.00 am | 37.5% | 30 min. |
| | 11.30 am | 50.0% | 30 min. |
| | 12.00 noon | 62.5% | 30 min. |
| | 12.30 pm | 75.0% | 30 min. |
| | 1.00 pm | 87.5% | 30 min. |
| | 1.30 pm | 100.0% | 24 hrs. |
| | | | |
| | 2nd Day | | |
| | 1.30 pm | 87.5% | 30 min. |
| | 2.00 pm | 75.0% | 30 min. |
| | 2.30 pm | 62.5% | 30 min. |
| | 3.00 pm | 50.0% | 30 min. |
| | 3.30 pm | 37.5% | 30 min. |
| | 4.00 pm | 25.0% | 30 min. |
| | 4.30 pm | 12.5% | 30 min. |
| | 5.00 pm | 0.0% | 60 min. |
| | | | |
| 2 | 3rd Day | 0.0% | |
| | 8.30 am | 25.0% | 30 min. |
| | 9.00 am | 50.0% | 30 min. |
| | 9.30am | 75.0% | 30 min. |
| | 10.00 am | 100.0% | 30 min. |
| | 10.30 am | 112.5% | 30 min. |
| | 11.00 am | 125.0% | 30 min. |
| | 11.30 am | 137.5% | 30 min. |
| | 12.00 noon | 150.0% | 30 min. |
| | 12.30 pm | 162.5% | 30 min. |
| | 1.00 pm | 175.0% | 30 min. |
| | 1.30 pm | 187.5% | 30 min. |
| | 2.00 pm | 200.0% | 24 hrs. |
| | | | |
| | 4th Day | | |
| | 2.00 pm | 175% | 30 min. |
| | 2.30 pm | 150.00% | 30 min. |
| | 3.00 pm | 125% | 30 min. |
| | 3.30 pm | 100.00% | 30 min. |
| | 4.00 pm | 75% | 30 min. |
| | 4.30 pm | 50.00% | 30 min. |
| | 5.00 pm | 25% | 30 min. |
| | 5.30 pm | 0% | 60 min. |

| Load Cycle No. | Day & Time | Load (% of WL) | Minimum Holding Time |
|---|---|---|---|
| 3 | 5th Day | 0.0% | |
| | 8.30 am | 25.0% | 30 min. |
| | 9.00 am | 50.0% | 30 min. |
| | 9.30am | 75.0% | 30 min. |
| | 10.00 am | 100.0% | 30 min. |
| | 10.30 am | 125.0% | 30 min. |
| | 11.00 am | 150.0% | 30 min. |
| | 11.30 am | 175.0% | 30 min. |
| | 12.00 noon | 200.0% | 30 min. |
| | 12.30 pm | 212.5% | 30 min. |
| | 1.00 pm | 225.0% | 30 min. |
| | 1.30 pm | 237.5% | 30 min. |
| | 2.00 pm | 250.0% | 30 min. |
| | 2.30 pm | 262.5% | 30 min. |
| | 3.00 pm | 275.0% | 30 min. |
| | 3.30 pm | 287.5% | 30 min. |
| | 4.00 pm | 300.0% | 24 hrs. |
| | | | |
| | 6th Day | | |
| | 4.00 pm | 250.0% | 30 min. |
| | 4.30 pm | 200.0% | 30 min. |
| | 5.00 pm | 150.0% | 30 min. |
| | 5.30 pm | 100.0% | 30 min. |
| | 6.00 pm | 50.0% | 30 min. |
| | 6.30 pm | 0.0% | 60 min, |
| | | | |

Following each application of load, hold the load for not less than the period shown or until the rate of settlement is less than 0.1mm/hr.

For any period when the load is constant, record time and settlement immediately on reaching the load and at 15-min. intervals for the 1st hour, at 30-min. intervals between the 1st and 4th hour and at 1-hr intervals between the 4th and 24th hour after the application of the increment of the load.

Continue testing until the maximum test load has been reached or when the settlement exceeds 10% of the pile diameter.

## 18.0 Document Control

18.1 Approval: GLOBAL_FAC_GFTT_APPR

18.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_ NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

18.3 Notification:

All Micron Team Members at all Sites

18.4 Retention

Adherence to the requirements specified in
**Global Facilities - Document Control Procedures**

18.5 Review

Biennially (two years)

## 19.0 Revision History

| Revision # | Section Changed | Details of Changes | Date |
|---|---|---|---|
| 0 | NA | New Document | 07/05/2022 |